

WASP-12b

R-0421

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_221226 · created 2026-07-19T08:37:06+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459939.87935 +/- 0.00097 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.111 +/- 0.0067
Transit depth (Rp/Rs)^2	1.23 +/- 0.15 [%]
Orbital Inclination (inc)	79.64 +/- 1.48
Airmass coefficient 1 (a1)	1.0161 +/- 0.0057
Airmass coefficient 2 (a2)	-0.0138 +/- 0.0055
Scatter in the residuals of the lightcurve fit is	0.56 %
Best Comparison Star	#2 - [322, 255]
Optimal Aperture	3.99
Optimal Annulus	11.76
Transit Duration (day)	0.114 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.111 ± 0.0067 vs 0.1178 (deviation 1.01σ)
Duration fit vs published	0.114 d vs 0.125 d (deviation 2.24σ)
Mid-time O–C	-1.1 min
Depth SNR	16.6
Residual scatter	0.56 %

Light curve

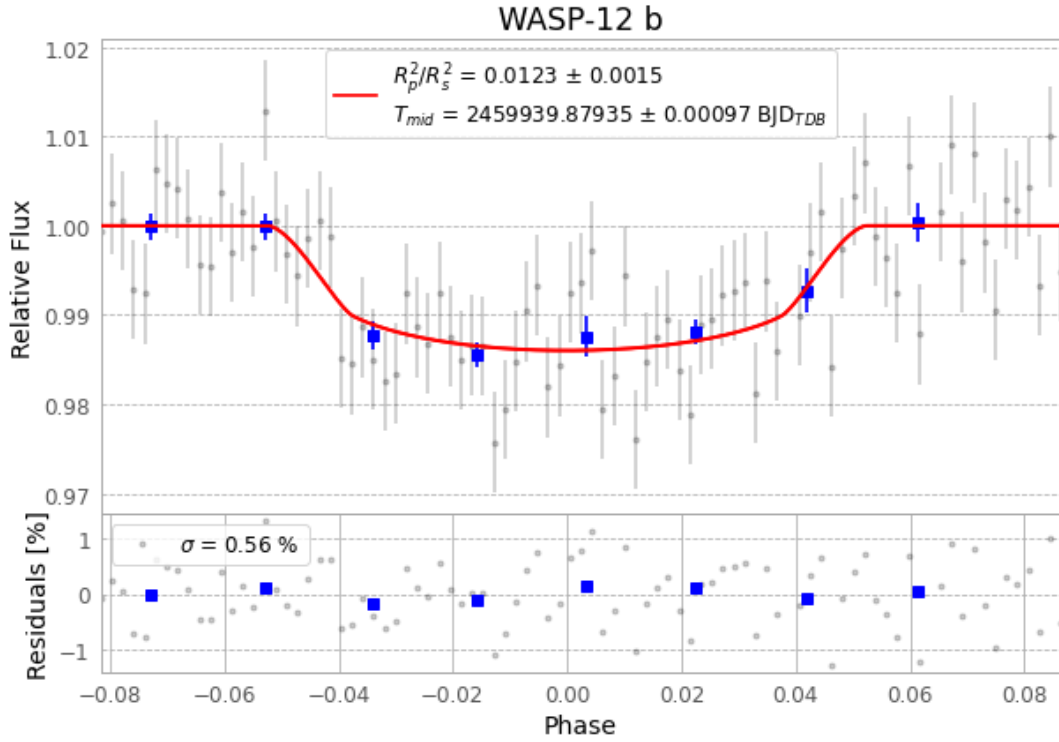


Figure 1 — FinalLightCurve_WASP-12 b_2022-12-25.png (SHA-256 efb8494999849d88...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [401, 217], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7556a168b5bc017a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

89 FITS frames (58 MB), WASP-12221226064815.FITS → WASP-12221226111215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-221226.log (6a015e2c2cc3...), FinalLightCurve_WASP-12 b_2022-12-25.png (efb849499984...), FinalLightCurve_WASP-12 b_2022-12-25.csv (1c7c3c14f4d1...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 08:37Z.